DevOps Lifecycle Diagram & Workflow Specification

Project: Smart Irrigation Alert Dashboard
Student: Chetan Katkar — **Roll No:** 23102A0024 — **Class:** BE CMPN-A

1. The 7 Stages of the DevOps Lifecycle (Mermaid Source)

```
flowchart TD
    subgraph SCM["1. Plan & Code"]
        A[Agile Backlog & User Stories] --> B[Feature Branching in Git]
        B --> C[Pull Request & Code Review on GitHub]
    end

    subgraph CI["2. Continuous Integration"]
        C -->|Webhook Trigger| D[Jenkins Pipeline Server]
        D --> E[Maven Clean Compile & Package]
        E --> F[Selenium WebDriver Automated Tests]
        F -->|Quality Gate Passed| G[Build Standalone Spring Boot JAR]
    end

    subgraph CD["3. Continuous Delivery & Containerization"]
        G --> H[Docker Build Multi-Stage Image]
        H --> I[Tag & Push to Docker Hub Registry]
    end

    subgraph Ops["4. Provisioning & Continuous Monitoring"]
        I --> J[Ansible Automated Provisioning]
        J --> K[Run Docker Container on Port 8080]
        K --> L[Health Check /api/sensors Endpoint]
        L --> M[Real-Time Dashboard & Threshold Monitoring]
        M -->|Feedback Loop| A
    end
```

2. Toolchain Matrix Across the Lifecycle

DevOps Stage	Tool Employed	Purpose in Smart Irrigation Project
1. Plan	Markdown / Kanban	User stories, sprint backlog, acceptance criteria
2. Code	Java 21, Spring Boot, Git	Source code development, modular branching
3. Build	Apache Maven 3.9	Dependency resolution, packaging executable .jar
4. Test	Selenium, JUnit 5	Browser UI automation, low-moisture alert validation
5. Release	Jenkins, GitHub	Declarative CI/CD pipeline as code
6. Deploy	Docker, Docker Hub	Container packaging, registry version tagging
7. Operate	Ansible, Actuator	Idempotent server provisioning, health-checks, metrics